

DAVID JOHN NIEDZWIECKI JR.

Principal AI Systems Architect & Backend Developer

Founder, SovereignNexus LLC

Location: New Bern, North Carolina

GitHub: github.com/BiggerRoarThnU/Geminiology

PROFESSIONAL SUMMARY

A highly technical, execution-focused backend developer and systems architect specializing in deterministic AI deployments, secure local API gateways, and robust data automation. Founder of SovereignNexus LLC, dedicated to building zero-trust digital infrastructure, orchestrating multi-agent Large Language Model (LLM) pipelines, and engineering air-gapped environments that prioritize data sovereignty, compliance, and operational efficiency.

CORE COMPETENCIES

- **Languages & Environments:** Python, Bash/Shell Scripting, Linux (Ubuntu/Debian), Git.
- **AI & Machine Learning:** Local LLM Deployment (Ollama, vLLM, Qwen, Gemma, Llama), Prompt Filtration, Multi-Agent Orchestration.
- **Backend Engineering:** FastAPI, REST API Development, JSON-RPC 2.0, System Telemetry, SQLite, Data Automation.
- **System Architecture:** Zero-Trust Security, Automated Rsync Backup, Hardware/Thermal Telemetry Daemons.

PROFESSIONAL EXPERIENCE

SovereignNexus LLC - New Bern, NC

Principal AI Systems Architect & Founder | March 2026 - Present

- **AI Architecture & Local Deployment:** Designed and implemented the "Geminiology" framework, a multi-agent AI environment running completely air-gapped on local edge hardware substrates.
- **Security & Data Sovereignty:** Engineered prompt-injection defense layers and middleware proxies (sovereign_defense_core.py) that sanitize inputs using regex and check outputs via vector similarity.
- **Backend API Integration:** Built custom Python/FastAPI wrappers (oracle_server.py) to wrap local model execution (Ollama) in a REST API, enabling secure internal query processing with zero external API fees.
- **System Automation & Health:** Developed metabolic governor loops (sovereign_loop.py) and custom telemetry tools (tools.py) to monitor CPU, RAM, and thermal thresholds, auto-throttling active runs to protect hardware.
- **Log & Data Processing:** Engineered high-throughput rsync backup scripts (master_t7_sync.sh) to mirror 8.5+ GB of files and logs to external solid-state storage.

EDUCATION & CERTIFICATIONS

- **Google AI Professional Tracks:** Local open-source model optimization, prompt engineering, and API gateways.
- **Coursera AI Fundamentals & Python Programming:** Modules in data processing, database management, and scripting.